

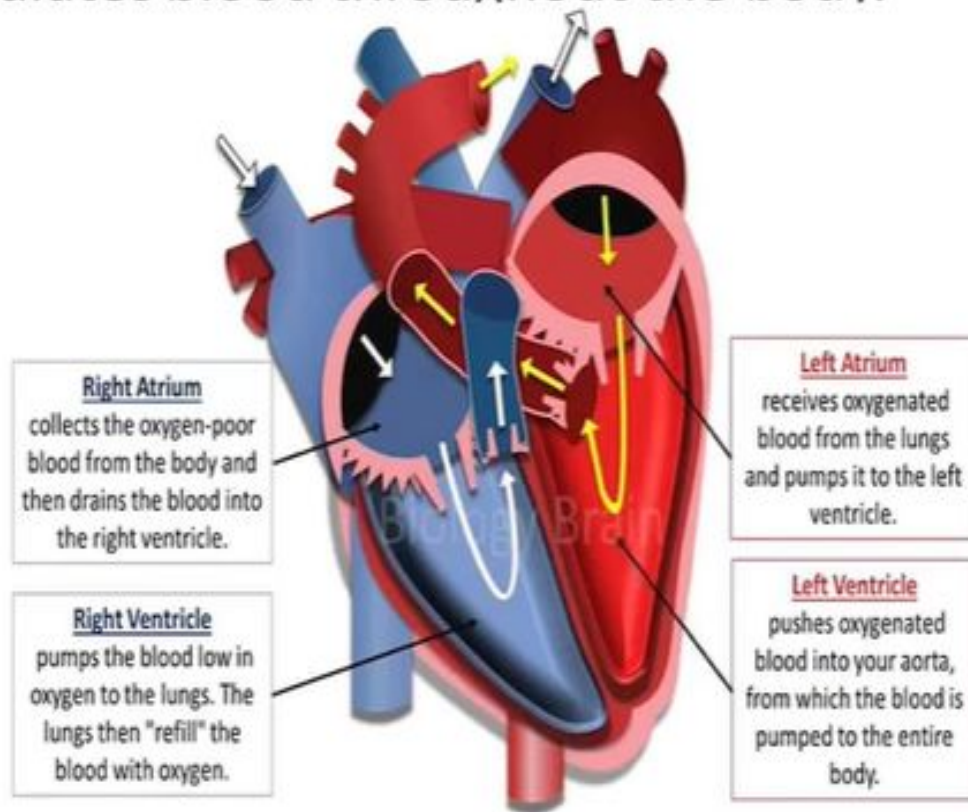
HEART AS A PUMPING SYSTEM



Heart as a Pump System

3.1 Architecture

- The heart is a complex pump system that circulates blood throughout the body.



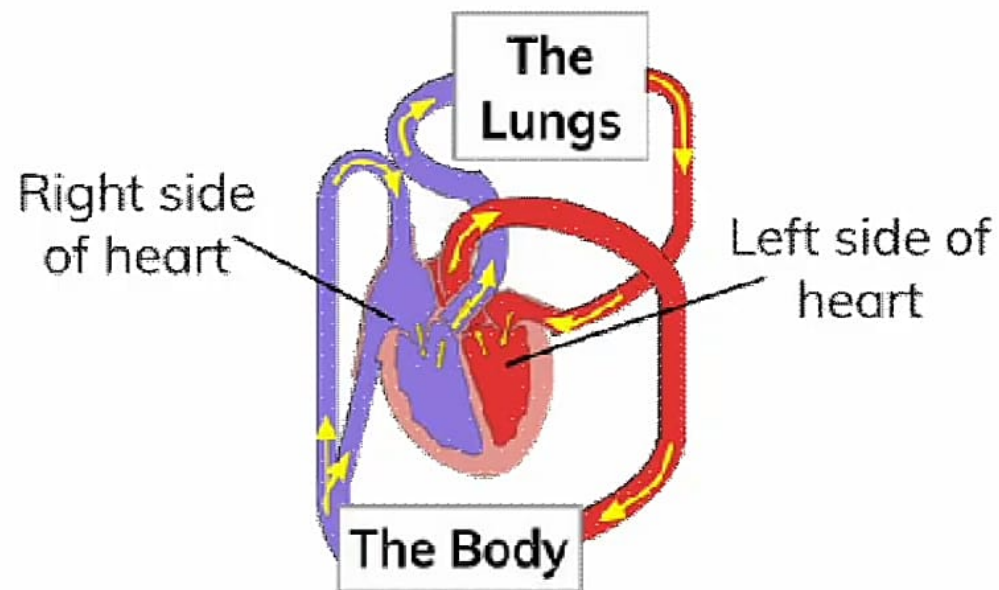


Figure 2.17: Representing circulation of blood

The Heart Beat

- The heart's pumping action is controlled by a **complex network of electrical and chemical signals**, which generate **the rhythm** of the heartbeat.
- An electrical stimulus is generated in a special part of the heart muscle called the **sinus node**. It's also called the sinoatrial node (SA node).
- The sinus node is a **small mass of special tissue** in the right upper chamber of the heart (right atrium).
- In an adult, the sinus node **sends out** a regular **electrical pulse 60 to 100 times per minute**.

- This electrical pulse **travels down** through the conduction pathways and **causes the heart's lower chambers (ventricles) to contract** and **pump out blood**.
- The right and left atria are stimulated first and contract to **push blood** from the atria into the **ventricles**.
- The ventricles then contract **to push blood out into the blood vessels** of the body

Electrical Signalling – ECG Monitoring and Heart Related Issues

- The heart's pumping action is controlled by **electrical signaling**, which generates the rhythm of the heartbeat.
- This electrical signaling can be monitored using an **electrocardiogram (ECG)**, which **records the electrical activity of the heart** and provides important information about the heart's function.

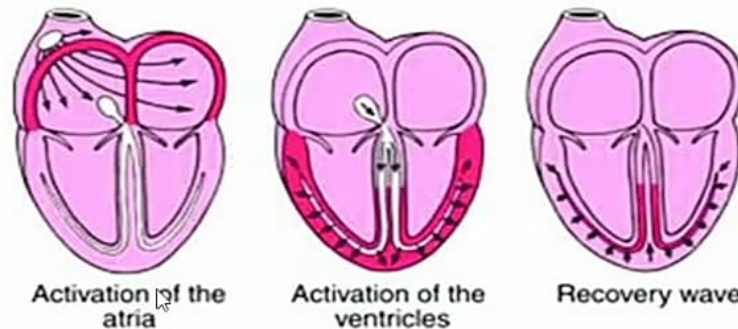
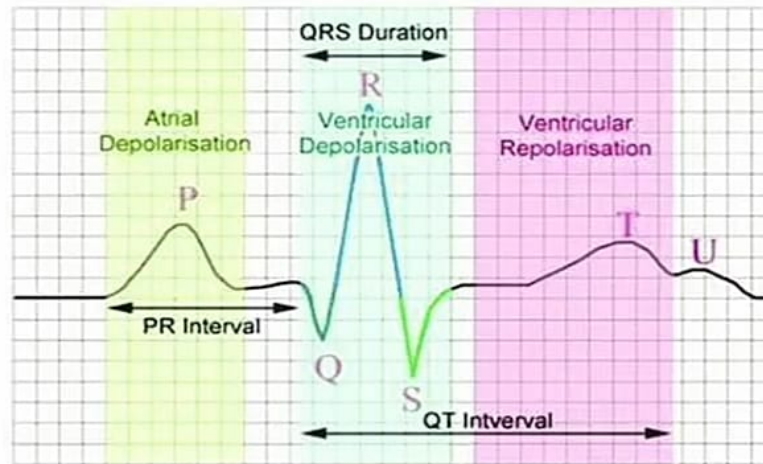
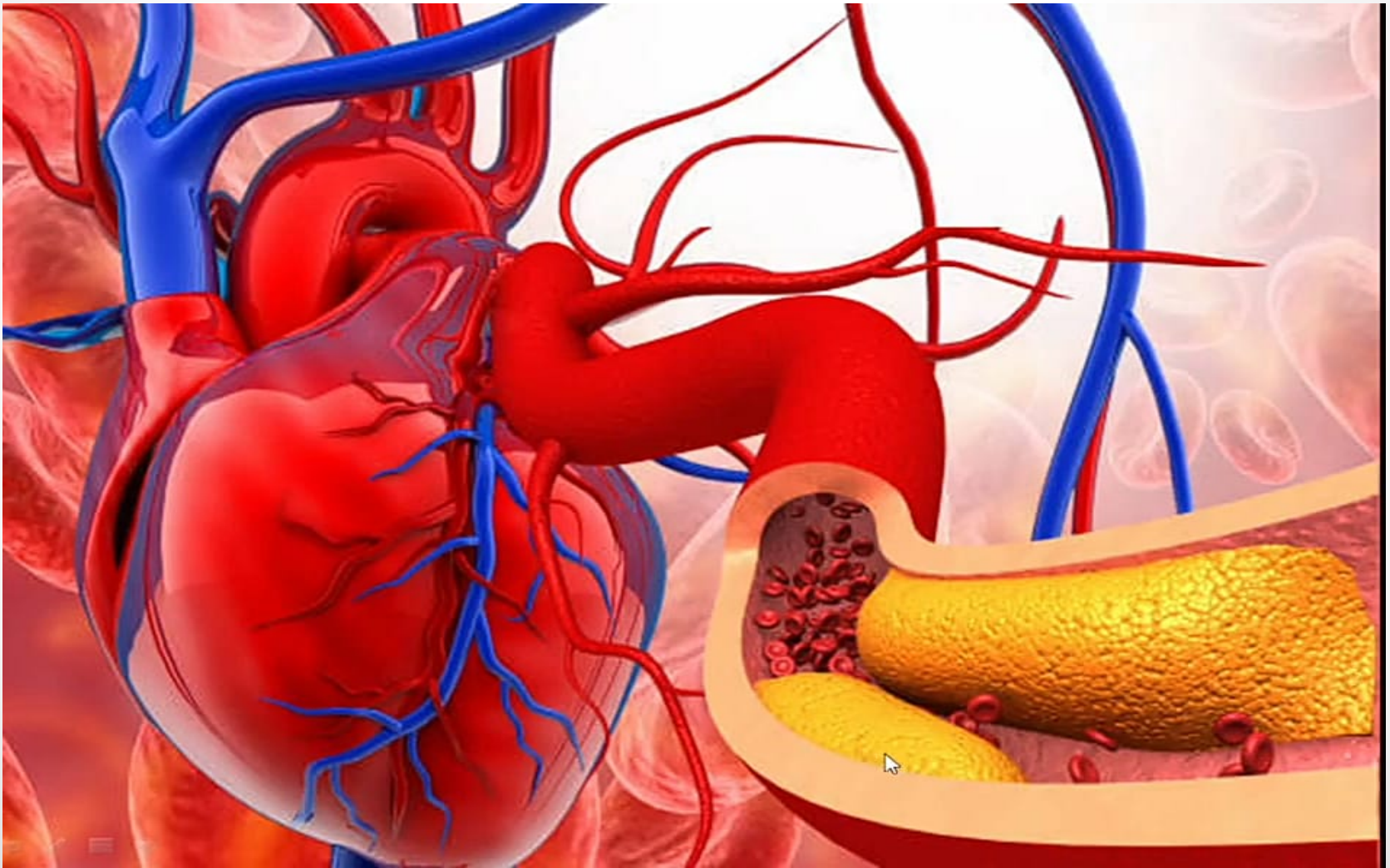


Figure 2.18: ECG waves and their relation to heart nodes

- Some common heart-related issues that can be diagnosed or monitored using an ECG include:
 - **Arrhythmias:** Abnormalities in the heart's rhythm or rate can be detected using an ECG.
 - **Heart disease:** Changes in the heart's electrical activity can indicate the presence of heart disease, such as coronary artery disease or heart attacks.
 - **Heart attack:** An ECG can help diagnose a heart attack by detecting changes in the heart's electrical activity that indicate a lack of blood flow to the heart.
- Overall, the ECG is a **useful tool** for **diagnosing and monitoring** heart-related issues and helps to provide important information about the heart's function and health.

BLOCKAGES OF BLOOD VESSELS



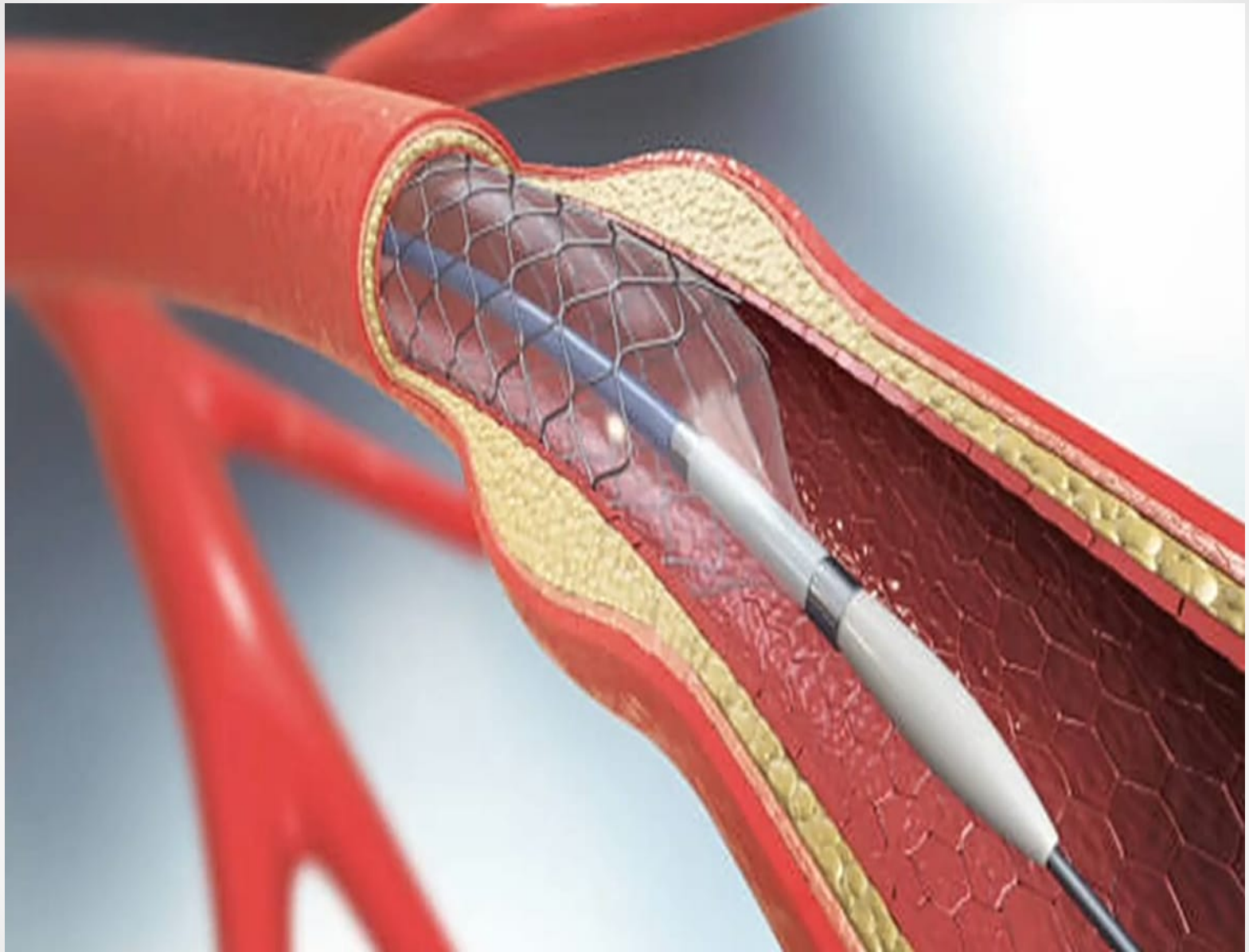
Reasons for Blockages of Blood Vessels

- Blockages in blood vessels, also known as arterial blockages or atherosclerosis, can occur for several reasons:
 1. **High cholesterol levels:** Excessive amounts of low-density lipoprotein (LDL) cholesterol in the blood can lead to the formation of plaque in the blood vessels, which can narrow or block them.
 2. **High blood pressure:** Over time, high blood pressure can cause damage to the blood vessels, leading to the formation of plaque and blockages.
 3. **Smoking:** Smoking can damage the inner walls of blood vessels and promote the buildup of plaque, leading to blockages.

4. **Poor diet:** A diet high in saturated fats, trans fats, and cholesterol can increase the risk of developing blockages in the blood vessels.
5. **Diabetes:** People with uncontrolled diabetes are at a higher risk of developing blockages in their blood vessels, due to damage to the blood vessels from high levels of glucose.
6. **Age:** As people age, the blood vessels can become stiff and less flexible, increasing the risk of blockages.
7. **Genetics:** Some people may be predisposed to developing blockages in their blood vessels due to genetic factors.

Design of Stents

- Stents are small, **metal mesh devices** that are **used to treat blockages** in blood vessels.
- They are typically used in procedures such as **angioplasty**, where a **balloon catheter** is used to **open up a blocked blood vessel** and a **stent** is placed to keep it open.



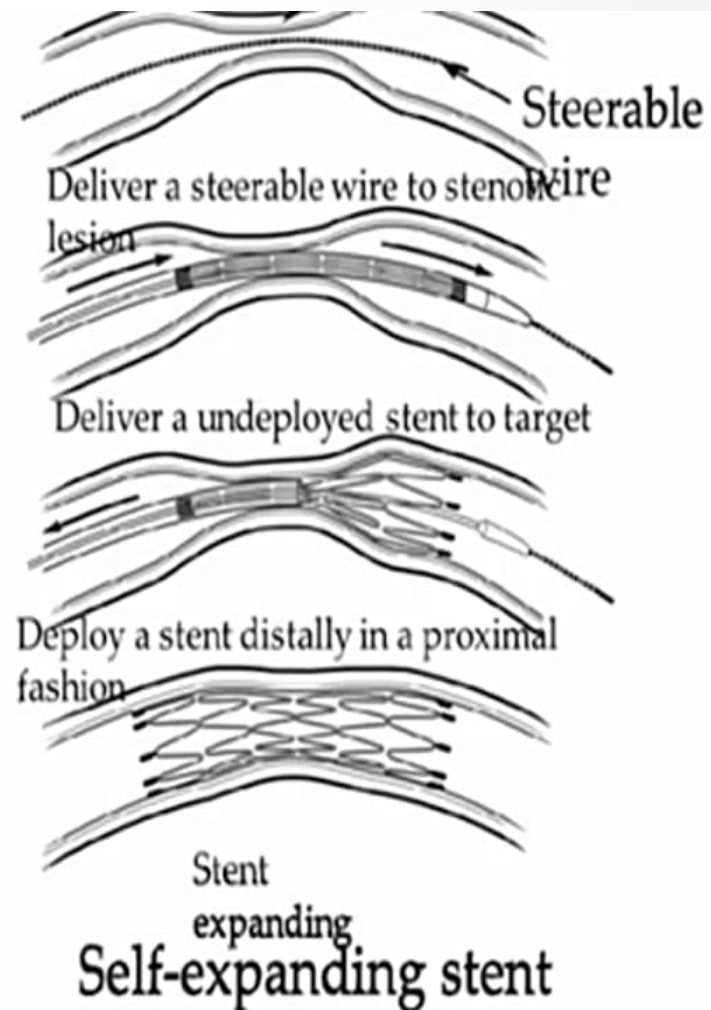
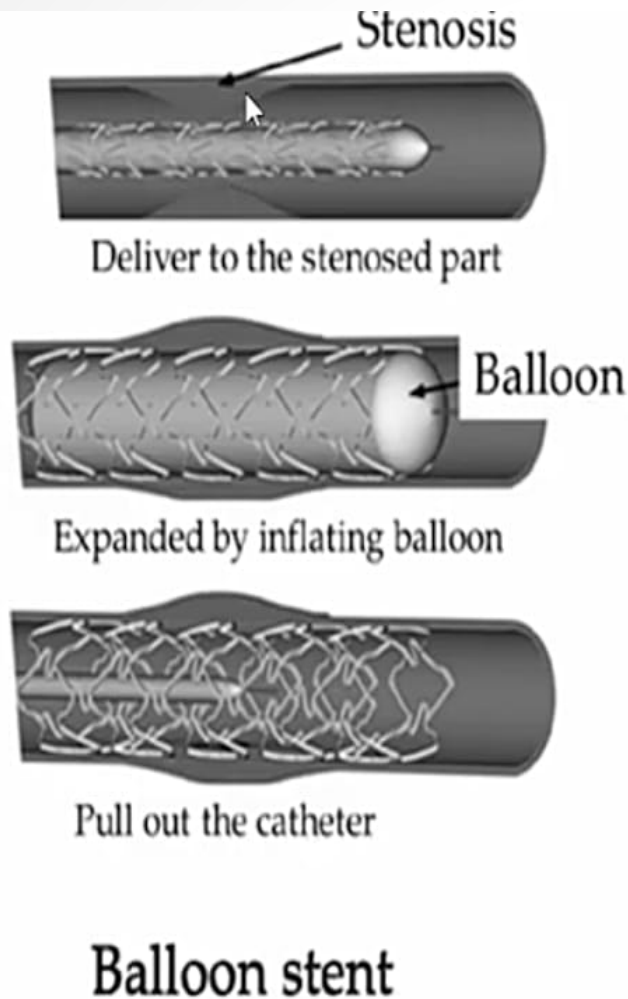


Figure Representing the working of balloon stent

Pace Makers

- A pacemaker is a small device that is **surgically implanted in the chest to regulate the heartbeat.**
- It is used **to treat heart rhythm disorders**, such as **bradycardia** (a slow heartbeat) or **arrhythmias** (abnormal heart rhythms), by delivering electrical impulses to the heart to regulate its rhythm.

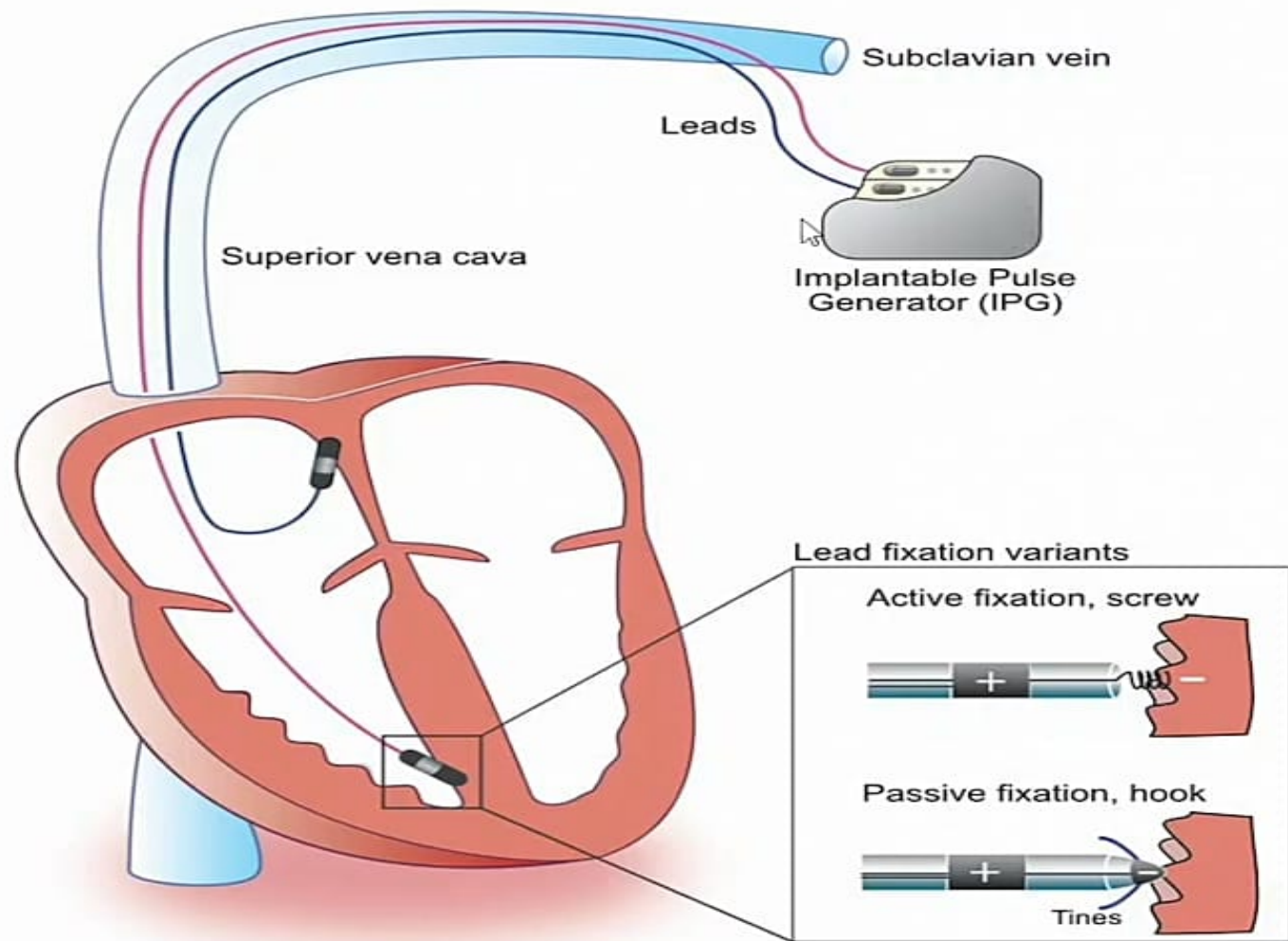
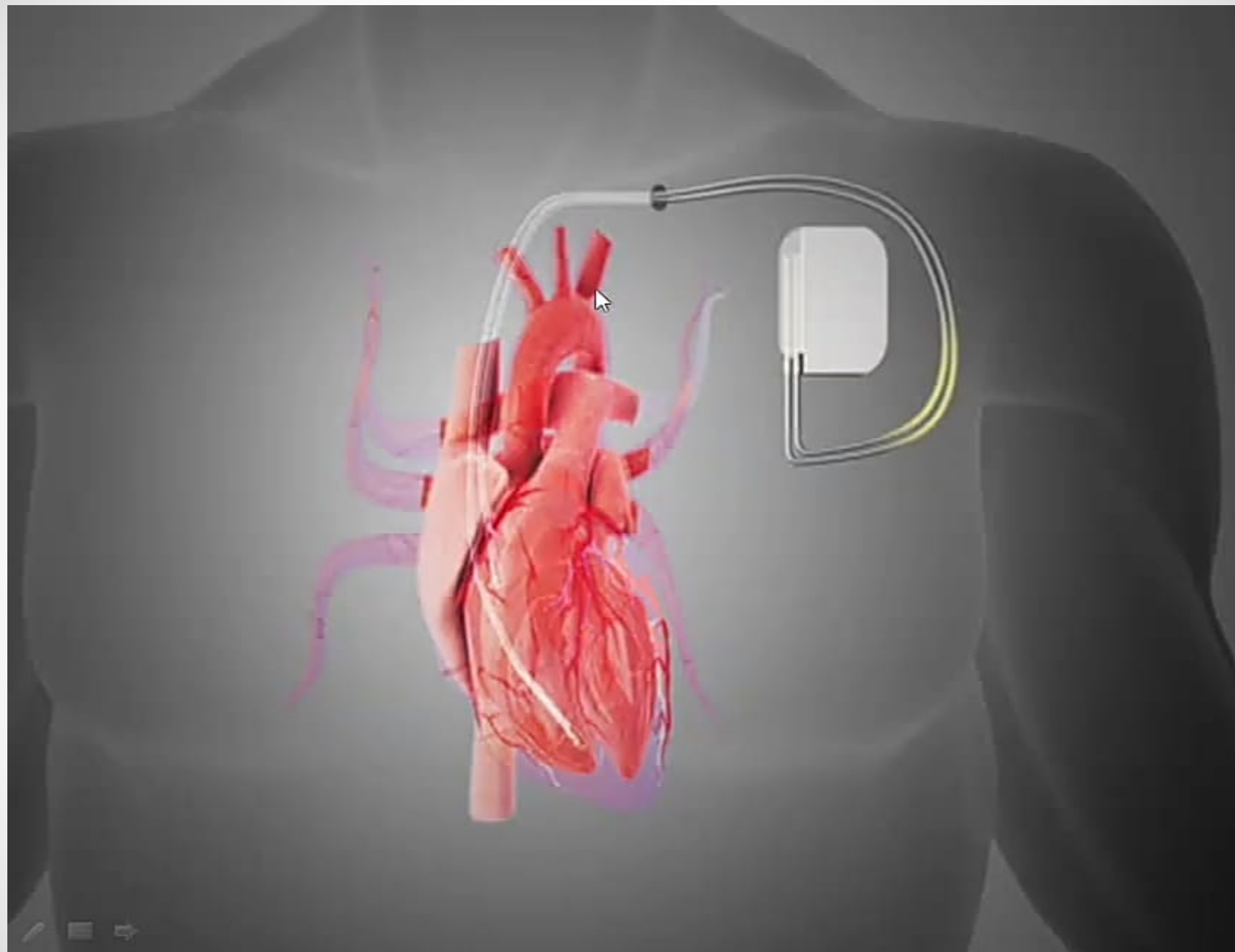


Figure []: Representing components of a pacemaker

The basic design of a pacemaker consists of:

- **Generator:** The generator is the main component of the pacemaker and contains a battery and electronic circuitry to generate and control the electrical impulses
- **Leads:** Leads are thin wires that connect the generator to the heart and carry the electrical impulses from the generator to the heart
- **Electrodes:** The electrodes are located at the end of the leads and are used to deliver the electrical impulses to the heart



Pacemakers can be designed to work in different ways, including:

- **Single-chamber pacemaker:** A single-chamber pacemaker delivers electrical impulses to either the right atrium or the right ventricle of the heart to regulate its rhythm
- **Dual-chamber pacemaker:** A dual-chamber pacemaker delivers electrical impulses to both the right atrium and the right ventricle of the heart to regulate its rhythm
- **Biventricular pacemaker:** A biventricular pacemaker delivers electrical impulses **to both ventricles of the heart to coordinate** their contractions and improve heart function in people with heart failure

Pacemaker Insertion

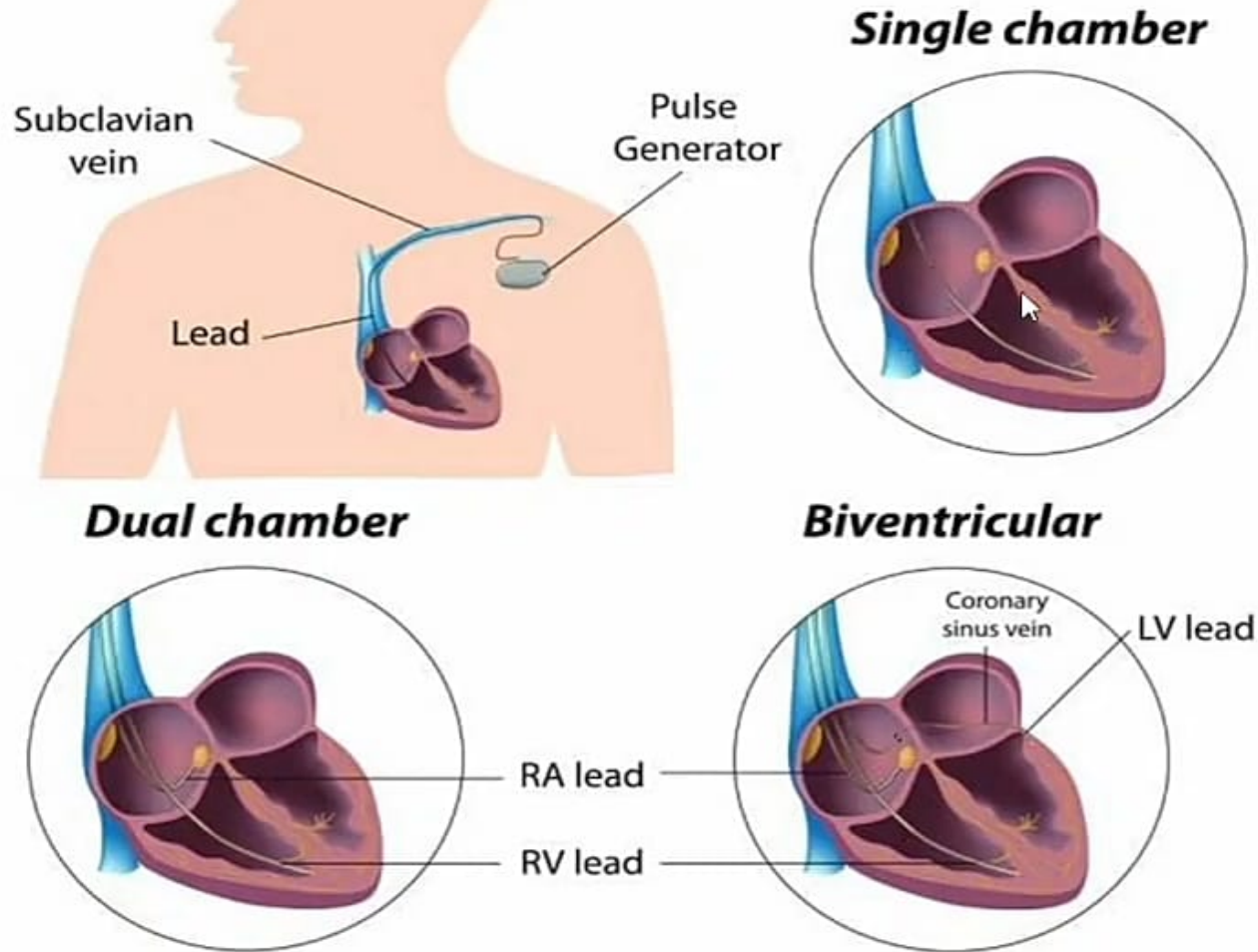


Figure 2.21: Representing the different types of pacemakers

Construction of a Pacemaker

- The construction of a pacemaker involves the use of **high-quality materials** and **specialized manufacturing processes** to ensure their **safety and reliability**.
- Materials used in the construction of pacemakers include:
 1. Medical-grade plastics: Medical-grade plastics, such as polycarbonate, are used to construct the exterior of the device and to provide insulation and protection for the internal components.

2. **Metals:** Metals, such as stainless steel and titanium, are used in the construction of the leads and electrodes to ensure their durability and long-lasting performance.
3. **Electronic components:** Electronic components, such as microprocessors, batteries, and capacitors, are used to control the delivery of the electrical impulses and to provide power to the device.
4. **Adhesives:** Adhesives, such as cyanoacrylate and epoxy, are used to secure the components of the device and to provide insulation and protection for the internal components.

- The manufacturing process for pacemakers includes **multiple quality control measures** to ensure their safety and reliability.
- This includes testing of individual components and final assembly testing to verify the proper operation of the device before it is released for use

Defibrillators

- A defibrillator is a medical device that **delivers an electric shock to the heart** to restore its normal rhythm in cases of cardiac arrest or other life-threatening heart rhythm disorders.
- Defibrillators **can be external** (placed on the chest) **or internal** (implanted within the body).

The basic design of a defibrillator consists of:

- **Power source:** The power source, typically a battery, provides energy to deliver the electric shock to the heart
- **Electrodes:** The electrodes are placed on the chest and deliver the electric shock to the heart
- **Circuitry:** The circuitry in the defibrillator controls the delivery of the electric shock, including the timing, strength, and duration of the shock
- **Display:** A display on the defibrillator provides information about the heart rhythm, battery life, and other relevant information